Week-1 test- final

1. List and explain the main features of Java.
   Main Feature of Java :
   - it is a plateform-independent language while it runs anyhwere in different OS
   - it is a OOP while it is based on Objects
   - Simple and secure language
   - it is Robust language,strong and capable of handeling error
   - it is a multitreaded language means runs many task together
   - It is a portable while work on any system
2. What is the difference between Java SE, EE, and ME in terms of release history?
   -Java SE ->Standard Edition for core java ,basic programing,it is firstly relase and main version
   -java EE -> Enterprice Edition ,now called Jakarta EE,used for enterprices application and web apps
   -Java ME -> Micro edition , used for mobile application and embeded devices
3. Why is Java called a platform-independent language?
   - Java code compiled into byteCode
   - That Bytecode runs on JVM
   - while JVM is available on different Operating System

4. How has Java evolved from JDK 1.0 to the latest version?
   JDK 1.0 ->oops and JVM ,basic feature

   - it introduce the concept of WORA(Write once run everywhere)
   - it provided the JVM for platform independence

   Java 5 -> added genrics,autoboxing ,Annotation
   Java 8 ->major update like Lambda Expression, stream Api
   Java 9 ->intro of module System
   Java 11 ->LTS ,log term support
   Java 17 ->new feature, better Security(LTS)
   Java 21+ ->improved Performance, Virtual threads
5. What makes Java "Secure"?
   - no Pointer Access (no direect memory access, used refernced)
   - program Runs Inside a JVM instead of OS
6. What are the different editions of Java?

- **Java SE (Standard Edition)** → Basic Java programming -**Java SE** is used for:
- Writing **basic Java programs**
- Building **desktop applications**
- Learning core concepts like:
  - Classes & Objects

    - Loops & Conditions
    - OOP concepts
- **Java EE (Enterprise Edition)** → Web and enterprise applications *(now called Jakarta EE)*
  Java Enterprize Edition is now called Jakarta Enterprise Edition it is used to build a large scalable web and enterprise apllications
  it includes
    - JSP-> Java server pages
    - Servlet -for handling the request
    - JPA -Java persistence API-Database handling
    - REST API-web services
    - EJB -Enterprize Java Beans
- **Java ME (Micro Edition)** → Mobile and embedded devices
  Stand for Micro Edition
    - it is used for Mobile Devices and Embeded devices like Tvs, Iot
    - it designs for small devices with limited memory

6. What is the difference between Java SE and Java EE?
   Java SE-

    - it is used to making a basic java apps
    - it includes a core liberaries like packages ex- java.util, java.lang, java.io
    - Desktop apps

Java EE (Now it is called Jakarta Edition)

- it is used to make a enterprise and web Application
- it includes advanced API like servlet,Websocket
- ex- Banking system and websites

7. What tools or APIs are included in Java EE?
   In Java EE- jakarta , main Api are:
   - Servlet Api - for hanling the request
   - JSP-Java server Pages, used to make a dynamic web pages-using server-side technology
   - JPA- Database handling
   - EJB-Enterprise Java beans for buisness logic
   - JMS -Java messaging system
8. What happened to Java EE when Oracle handed it over to Eclipse Foundation?

- Name changed to **Jakarta EE**
- Development became open-source
- Community now manages it

7. Difference Between C, C++, and Java
   - C --
     -procedual type
     -memory management is manual
     -platform dependent
     -pointer based
   - C++ --
     -Object-oriented

-manual memory
-platform dependent
-pointer based

- Java --
  -Object-oriented
  -Automatic memory-GC
  -platform independent
  -not a pointer based

8. Is Java fully object-oriented? Explain.
   not a 100% but supports OOPs
   - bcz of used a primitive data types like int, char, etc it is not become a 100% OOps lang
9. How does Java handle memory differently from C++?
   -C++ used manual memory management menas it manualy by using new for crate ,and delete for clean

   -java , concist of GC - automatic garbage collector
10. Why doesn't Java support pointers? ---> Direct memory access in c++
    - for security reasons
    - avoid by memory error
    - make a code simpler and safer
11. How is exception handling different in C++ and Java?
    -C++ - Use try-catch but not Strict
    -Java - Strict(must handle)/java has Checked Exception
12. Why does Java not support operator overloading?
    👉 To keep Java:

- Simple
- Easy to understand

  👉 Example:
- `+` works only for numbers & strings
- You cannot change its behavior

11. 🔹 **Access Specifiers (C++ vs Java)**

Feature	C++	Java
Types	private, public, protected	private, public, protected, default
Default access	private	package-private (default)

12. What is the difference in memory management between C++ and Java?

🔹 **Direct Memory Access**

👉 C/C++ → Yes (pointers)
👉 Java → No ❌
👉 Java uses **references** for safety

13. Can you directly manage memory in Java like in C/C++?
14. Types of Java Applications

🔹 **Main Types**

1️⃣ Standalone Application

👉 Desktop / console-based programs
📦 Examples:

- Calculator
- Notepad
- Your Even/Odd program

2️⃣ Web Application

👉 Runs on server, accessed via browser
📦 Example:

- Login system
- Online shopping site
  👉 Technologies:
- Spring Boot
- Servlet
- JSP

3️⃣ Enterprise Application

👉 Large-scale apps for businesses
📦 Examples:

- Banking system

- ERP systems

  👉 Technologies:

- Spring Framework
- Hibernate
- Spring Boot

4️⃣ Mobile Application

👉 Apps for mobile devices
📦 Example:

- Android apps

  👉 Uses:

- Java (Android SDK)

5️⃣ Cloud / Distributed Application

👉 Apps running on multiple systems (servers)
👉 Java supports via:

- Spring Boot
- Microservices
- REST APIs

15. What are the different types of applications that can be developed using Java?
16. What is a standalone application? Give examples.

17. What is a web application in Java?
18. What technologies are used to create web applications in Java?
19. What is an enterprise application?
20. Give examples of enterprise-level Java technologies.
21. What is a mobile application in Java?
22. How does Java support cloud or distributed applications?
23. Which Java edition is used for embedded systems?

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concept-2 , JDK -**JDK (Java Development Kit)**

```
├── JRE
│   ├── JVM
│   └── Libraries (API)
└── Development Tools (javac, jar, javadoc)
```

JRE -(Java Runtime Environment)

```
├── JVM
└── Core Libraries (java.lang, java.util, etc.)
```

JVM -(Java Virtual Machine)

JVM

```
├── Class Loader
├── Runtime Data Areas
│   ├── Method Area
│   ├── Heap
```

```
│  ├── Stack
│  └── PC Register
├── Execution Engine
│  ├── Interpreter
│  └── JIT Compiler
└── Garbage Collector
```

1. What is the difference between JDK, JRE, and JVM?

- JDK- it is a software package used to devlop and run java program.
  -tools are part of JDK
- Javac- compiler
- JDB- debuguer
- javadoc- generate documentation
- jar - generate jar files
- JRE- it run java program only
  -JRE included :
    - JVM- Java virtual Machine -it execute java programs
    - core lib -like java.lang, java.util and so on
    - supporting files - Configs files
- JVM - execute java bytecode
  -JVM -> Java virtual Machine -- jvm is considered as most powerful bcz it allowing java application to be secure, self tuning, truly portable
- it is used to execute the program.

- its working is to (called engine) runs java bytecode

2. What are the common JDK commands and their purposes?
`javac`	Compile `.java` file
`java`	Run `.class` file
`jar`	Package files
`javadoc`	Create documentation
`jdb`	Debug program
3. What is the difference between Oracle JDK and OpenJDK?

- |Oracle JDK|
  -Commercial (some versions require license)
  -Provided by Oracle
- OpenJDK
  -Free and open-source
  -Community-supported

4. How does JDK help in development and compilation?
   JDK helps by:
   `.java → (javac) → .class → (JVM) → Output`

- Writing code in `.java`
- Using **javac** to compile → `.class` file
- JVM runs the bytecode
- Tools help debug, package, and document

5.JRE included :
JVM- Java virtual Machine -it execute java programs
core lib -like java.lang, java.util and so on
* supporting files - Configs files

6. How is JRE different from JDK?
    - JRE- it used to run and execute the program,it does not include compiler +jre is a small package
    - JDK - is used to run + develop the program ,it includes compiler +large package
7. What are the components of JRE?
    - JVM -used to execute the program
    - liberaries - java.util,java.lang
    - bytecode verifier is also there to check a code safety
8. How does JRE interact with JVM?
    JRE contain JVM , it is a core engine inside the JRE
    - jre provides lib like java.util, java.lang and the environment to run the program
    - JVM- it used to execute the code

Story of JVM

JVM->
consist of 4 parts

1.Class loader:
2.memory area
3.execution engine
4.GC

in class loader -> working : it loads the .class file into JVM and convert bytcode into memory
three types of class loader:
1.bootsrap class loader
2.application class loader
3.extenstion class loader

memory area : it provide memory to store the running program
memory parts :
1.method area -->store the class info,static variables
2.heap --> it stores the objects
3.stack -> local variables, methods calls
4.p register -> current instruction

Execution engine : final exectute the given bytcode and convert into machine code
components :
1.interpreter -> runs code line by line
2.JIT compiler -> used to make a faster for convert the bytecode into machine code

role of JVM : convert bytecode into machine code, makes the java platform independent

java program lifecycle: code write(.java file)--->compile by javac(compiler) make "bytecode(.class file)"---> JVM consist class loader which load it and execute the program.

component of JVM-- 1.Class Loader 2.memory access 3.execution Engine

1.Class Loader

- it is used to load the .class file compiled by the javac compiler
- working flow - Loading---->Linking ------>Intialization

2.memory access

- JVM automatically manage the memory
- it use a GC - garbage collector

3.execution Engine

- execute the bytecode(.class compiled by javac compiler) file using :
- Interpreter

- JIT compiler

  JVM Memory areas-- Heap|| Stack || Method Area || PC register

Heap area: - Stores objects
Method area - it store the class Info , static data

GC-it remove the unuses objects and automatically free the memory

- JVM nature is platform Indepenedent but Java is a platform dependent
- JVM - bytecode handeling by reads bytecode and convert to machine code.
- Native methods are written in C/C++.
- Each OS have its own JVM implementation

Class loader-- Bootstrap || Extension || Application

- Delegation model means parents load first if not found then child loads [parent first rule]
- Verfication Errors- occurs bcz of invalid code / security issues/ type mismatch
- linking phase : Verify/ prepare(allocate the memory for static variable)/ resolve

- Intialaztion phase: execute the Static variable and static blocks.

StackOverFlowError----> stack memory full
OutOfMemoryError ----> heap full

1. What is the role of JVM in Java execution?
2. Explain the lifecycle of a Java program in terms of JVM.
3. What are the main components of JVM?
4. What is the role of the Class Loader in JVM?
5. What is the role of the Execution Engine?
6. How does the JVM perform memory management?
7. What are the different memory areas in JVM?
8. What is method area and heap area?
9. How does JVM perform garbage collection?
10. Is JVM platform dependent or independent?
11. How does JVM handle bytecode?
12. What are native methods in JVM?
13. How does the JVM differ for different operating systems?
14. In which order does the JVM load classes (hierarchy of class loaders)?
15. What is the delegation model of class loading?
16. What kind of errors are detected during verification?
17. What is the Prepare phase and what happens during it?
18. What is the Resolution step in linking?

19. What is the main purpose of the initialization phase?
20. What happens when a StackOverflowError occurs?
21. What happens when an OutOfMemoryError occurs in the Heap?
22. What is the Native Method Stack, and why is it separate from JVM stack?

JIT - (Just-In-Time Compiler) Story

1. What is JIT compiler in Java?
2. Why is JIT compiler used?
3. When does JIT compile code?
4. How does JIT improve performance?
5. What is the difference between interpreter and JIT?
6. What are the types of JIT optimizations?
7. How does JIT work with bytecode and native code?
8. Is JIT part of JVM or JRE?
9. What happens if JIT is disabled?

10. Basic Steps to Write and Run a Java Program

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1. What is the extension of a Java source file?
2. What is the extension of the compiled bytecode file?

3. Which command is used to compile Java code?
4. Which command is used to run a Java program?
5. Why do we not use .class when running the program?
6. What is the entry point of a Java program?
7. What happens if the file name and class name differ?
8. What is bytecode?
9. Can we have multiple classes in one Java file?
10. What happens when we omit public keyword in class?
    Coding Questions
11. Write a program to print "Hello, Java Developer!"
12. Write a program to add two numbers
13. Write a program to find the largest of two numbers
14. Write a program to check even or odd number
15. Write a program to print sum of first 10 natural numbers
16. Write a program to swap two numbers
17. Write a program to swap two numbers without using third variable
18. Write a program to print ASCII value of a character
19. Write a program to check if number is positive or negative
20. Write a program to print multiplication table

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1. What is Object Class in Java?
   object class is a root of java class.

2. What is the superclass of all Java classes?
   obj class is a super class of all java class -> which means every class autonatically inherit obj class:
   package : java.lang.object
   it provide a common methods that every java obj can use
3. Is it necessary to extend Object class explicitly?
   No it is not required to extend object explicitly bcz java automatically object extend by all classes.
4. How many methods does the Object class have?
   there are 11 methods
   in which imp are
   - toString
   - equals()
   - == method
   - hashcode()
   - finalyze() [depriciated]
   - wait()
   - notify()
   - notifyall()
   - clone()
5. Can we override methods of the Object class?
   yes we can like toString(),hashcode(),eqauls. accept the final method like notifyAll(),wait()..
6. Which methods are declared as final in the Object class?
   methods which are declared as a final such as notify(),notifyAll(),wait().

7. What is the difference between == and equals()?
   basically == operator is used for compare the memory address,
   equals() is used for compare the content
8. **Why should we override hashCode() when we override equals()?**
9. What is the default implementation of toString()?
   class name + hashcode
10. What is the purpose of clone() method?
    clone method is a object class method , it is used to make a copy.
11. What is shallow copy?

- it create a copy of objects but not make copy of inner object
- it just copy the referneces
- it is dependent while making copy
- shared

12. What is deep copy?

- it crate the copy both the inner objects and external objects
- it is independent it does not affect any obj if we make a deep copy
- seperated

13. What is the purpose of finalize() method?
    it is depriciated method now no use
14. What is the use of getClass()?
    getClass() is used to return fully qualified className of the object .
15. Why are wait(), notify(), and notifyAll() methods defined in Object class and not in Thread class?
    these methods are not mention in a thread class bcz of certain reasion:
    1. we are using object as a lock for thread communication ,that means locking is happening in object not in thread ,
    2. example : a room is a object(lock) and people are thread ,we can control lock of room not people(threads)
16. What happens if we don't override toString()?
    it show a default value like classname+@hashcode ,which is a unreadable (not meaningfull)
17. Can we override finalize()?
    yes we can but not recommended (depriciated)
18. Can we call clone() without implementing Cloneable?
    no , we cant
    to call clone() implement cloneable
19. Why is finalize() deprecated in Java 9 and later?

    Coding Questions :
20. Write a Java program to display object details using toString() method.

21. Write a Java program to compare two objects using the equals() method.
22. Write a program to demonstrate the difference between == operator and equals() method.
23. Write a Java program to find whether two employee objects are equal by overriding equals() and hashCode().
24. Write a Java program to print the runtime class name using the getClass() method.
25. Write a program to clone an object using the clone() method.
26. Write a Java program to demonstrate deep cloning using the clone() method.
27. Write a Java program to override the hashCode() method.
28. Write a program to check the hash codes of two equal objects.
29. Write a Java program to call the finalize() method and see when it executes.
30. Write a Java program to demonstrate shallow copy?
31. Write a Java program to demonstrate deep copy?

----------end----------the ---------topic----------